

Experiment 'p_2_4_f_max' Results

June 3, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_2_4_f_max

Natural language statement: Write a method that takes two integers and then returns the maximum (the larger of the two).

Method signature: p_2_4_f_max(x: int, y: int) returns (result: int)

Ensures

- `result == x || result == y`
- `if x >= y then result == x else result == y`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that takes two integers and then returns the maximum (the larger of the
→ two).

The signature should be:

`p_2_4_f_max(x: int, y: int) returns (result: int)`

The method should respect the following contract:

`ensures result == x || result == y, ensures if x >= y then result == x else result == y`

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmdXZU22pP0LubprgaiqaNQJESL3r

Model: gpt-5.5-2026-04-23

Created at: 1780484109

Finish reason: stop

Usage: 304 tokens in, and 68 tokens out

```
{
  if x >= y {
    result := x;
  } else {
    result := y;
  }
}
```

Final Program

```
method p_2_4_f_max(x: int, y: int) returns (result: int)
  ensures result == x || result == y
  ensures if x >= y then result == x else result == y
{
  if x >= y {
    result := x;
  } else {
    result := y;
  }
}
```

```
}  
}
```

Total Token Usage

Input tokens: 304

Output tokens: 68

Reasoning tokens: 21

Sum of ‘total tokens’: 372

Experiment Timings

Overall Experiment started at 1780484108752, ended at 1780484115065, lasting 6313ms (6.31 seconds)

Iteration #1 started at 1780484108756, ended at 1780484115065, lasting 6309ms (6.31 seconds)